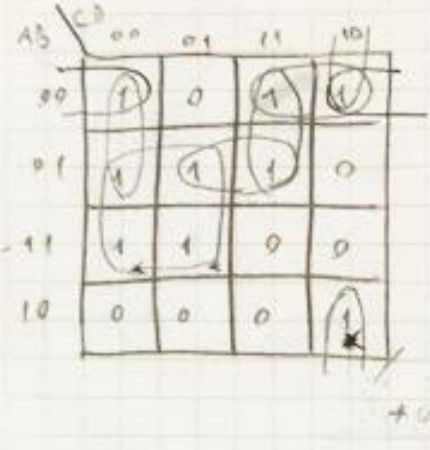
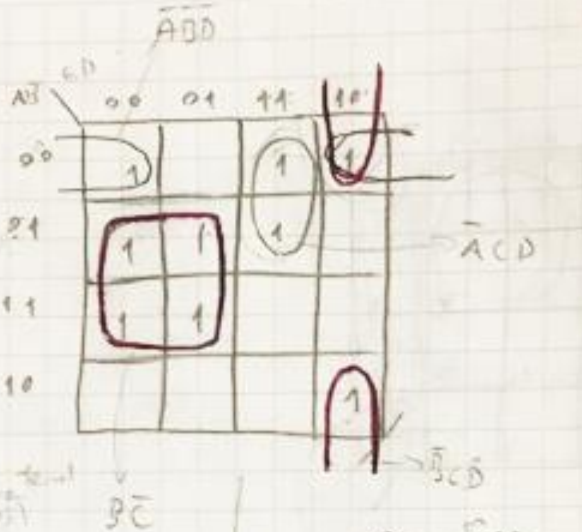


AB\CD	00	01	11	10
00	1	0	1	1
01	1	1	1	0
11	1	1	0	0
10	0	0	0	1



AB\CD	00	01	11	10
00	1		1	1
01	1	1	1	
11	1	1		
10				1

önce gerekli (EPI) bi yapıyoruz

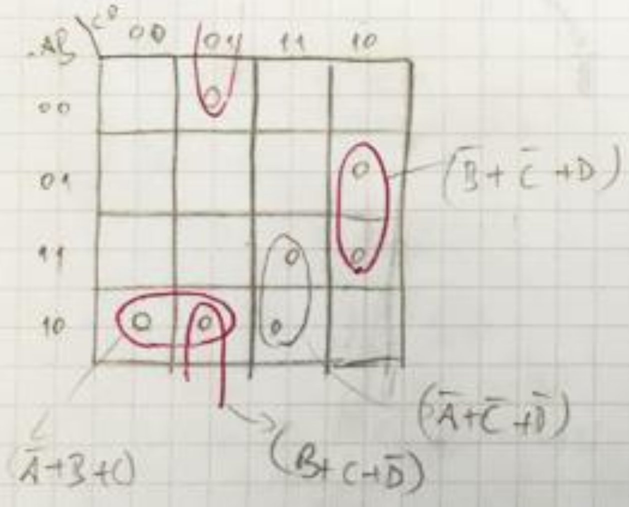
PI
• $B\bar{C}$
○ $\bar{A}\bar{C}\bar{D}$
• $\bar{A}\bar{B}\bar{D}$
○ $\bar{A}BD$
• $\bar{A}CD$
$\bar{A}\bar{B}C$
• $\bar{B}C\bar{D}$

EPI
$B\bar{C}$
$\bar{B}C\bar{D}$

OPI
$\bar{A}\bar{B}\bar{D}$
$\bar{A}CD$

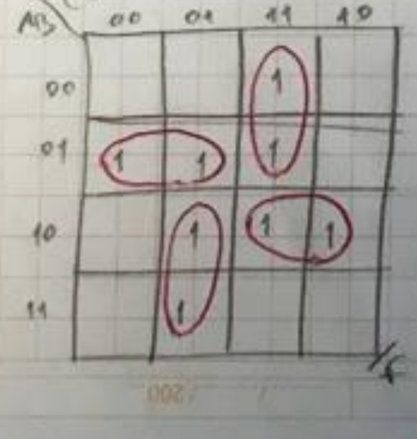
RPI
$\bar{A}\bar{C}\bar{D}$
$\bar{A}BD$
$\bar{A}\bar{B}C$

$$f = \underbrace{B\bar{C} + \bar{B}C\bar{D}}_{EPI} + \underbrace{\bar{A}CD + \bar{A}\bar{B}\bar{D}}_{OPI}$$



AB\CD	00	01	11	10
00		0		
01				0
11			0	0
10	0	0	0	

$$f_{pos} = (\bar{A}+B+C)\cdot(B+C+\bar{D})\cdot(\bar{A}+\bar{C}+\bar{D})\cdot(\bar{B}+\bar{C}+D)$$



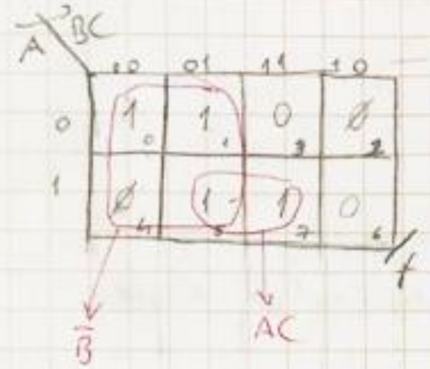
AB\CD	00	01	11	10
00			1	
01	1	1	1	
10		1	1	1
11		1		

* önce Mutlaka önce gerekliliklerden başlıyoruz bunu yaparken de Maksimum komşuluk ilişkisini gözetiyoruz

Incompletely specified functions
(DON'T CARE)

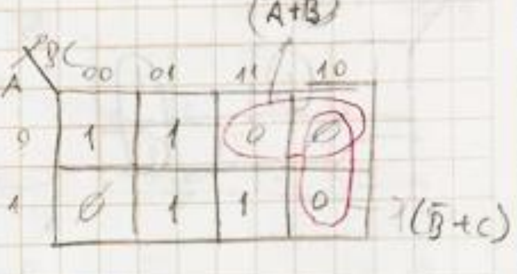
$f(A,B,C) = \Sigma m(0,1,5,7) + \emptyset(2,4)$

$g(A,B,C,D) = \Pi M(0,1,4,6,8,14,15) \cdot \emptyset(2,3,9)$

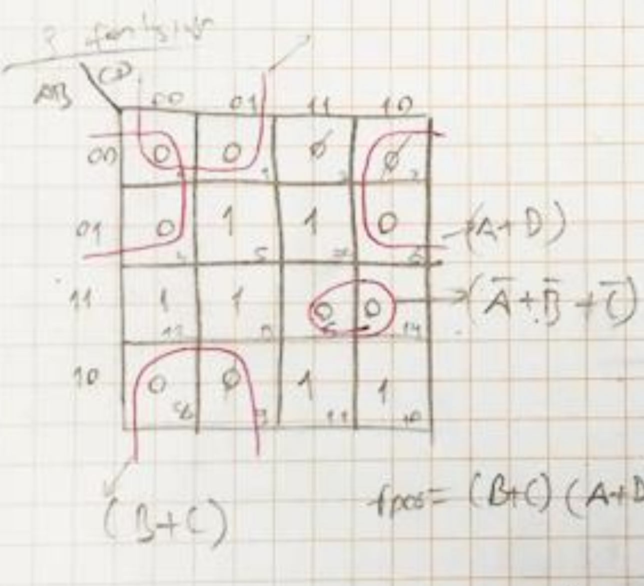


GRAY CODE

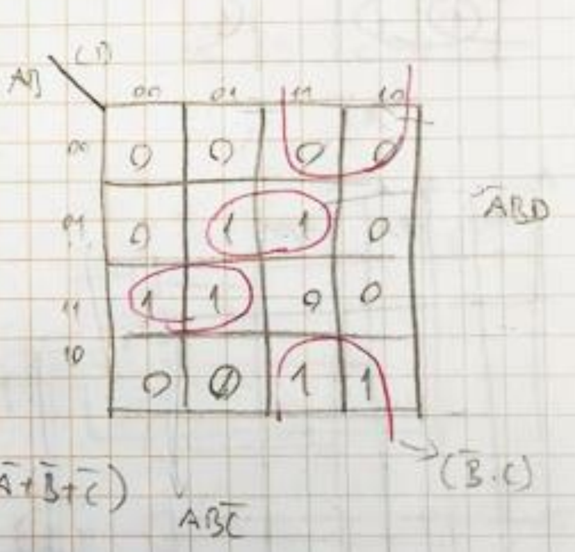
$f_{sop} = \bar{B} + AC$



$f_{pos} = (A+\bar{B}) \cdot (\bar{B}+C)$



$f_{pos} = (B+C)(A+D) \cdot (\bar{A}+\bar{B}+\bar{C})$



$f_{sop} = AB\bar{C} + \bar{B}C + \bar{A}BD$

ENTERED VARIABLE MAP REDUCTION
(MEV)

N : Input variable

n : order of MAP

$(n < N)$

$N-n$: free variable = Entered variable

2^{N-n} : Map key, Truth key

$$\begin{pmatrix}\text{Compressed Map} \\ \text{Cell Number}\end{pmatrix} \cdot (\text{MAP Key}) = \begin{pmatrix}\text{Code Number of the first Minterm or} \\ \text{Maxterm possible for the} \\ \text{compressed Map cell}\end{pmatrix}$$

$f(A,B,C) = \sum m(2,5,6,7)$

	Dec	A (MSB)	B	C (LSB)	f	
0	0	0	0	0	0	
	1	0	0	1	0	0
1	2	0	1	0	1	$\overline{C}$
	3	0	1	1	0	
2	4	1	0	0	0	C
	5	1	0	1	1	
3	6	1	1	0	1	1
	7	1	1	1	1	

$N=3$, $n=2$ ⟩ $N-n = 3-2 = 1$

$2^{N-n} = 2^1 = 2$

LSB

En düşük ağırlıklı değişken serbest değişken olarak seçilir

free variable : C

$0 \cdot 2 = 0 \rightarrow 0,1$

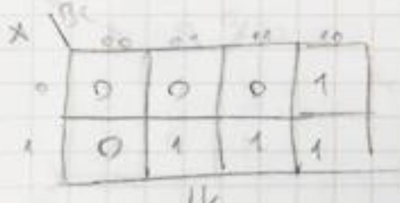
$1 \cdot 2 = 2 \rightarrow 2,3$

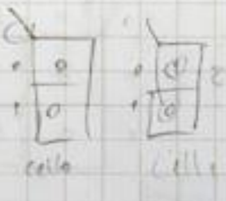
$2 \cdot 2 = 4 \rightarrow 4,5$

$3 \cdot 2 = 6 \rightarrow 6,7$

⇓

Dec	A	B	f
0	0	0	0
1	0	1	$\overline{C}$
2	1	0	C
3	1	1	1

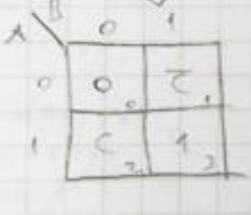




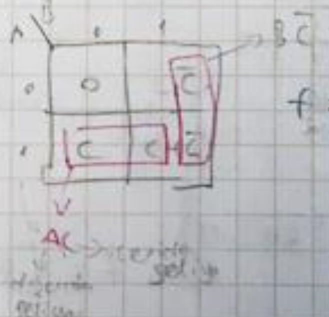
$C + \overline{C} = 1$

$C \cdot \overline{C} = 0$

⇓

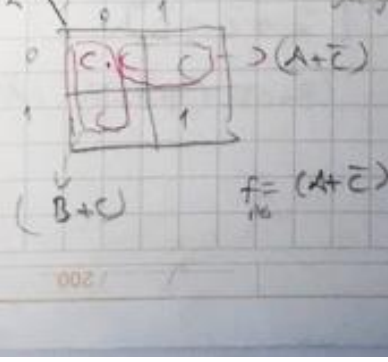


⇒



$f = B\overline{C} + AC$

⇓



$f_{\text{POS}} = (A+\overline{C}) \cdot (B+C)$

ör: $f(A,B,C,D)=\Pi M(0,1,6,8,9,11,14,15)$

AB \ CD	00	01	11	10
00	0_0	0_1	1_3	1_2
01	1_4	1_5	1_7	0_6
11	1_{12}	1_{13}	0_{15}	0_{14}
10	0_8	0_9	0_{11}	1_{10}

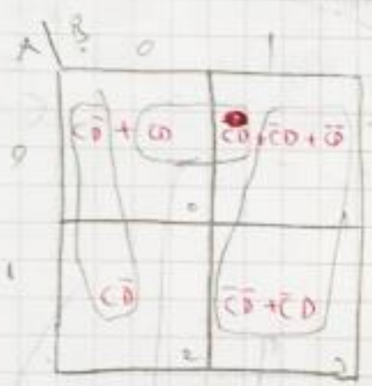
→ 2. mertebeden karnou haritasın dönüştürün

$N=4$

$n=2$

$N-n=2$ (free variable)
C D

$2^{N-n}=2^2=4$



A
B
0
1
0
1
$C\bar{D}+CD$
$\bar{C}\bar{D}, \bar{C}D+\bar{C}\bar{D}$
$C\bar{D}$
$\bar{C}\bar{D}+\bar{C}D$
0
1
2
3

$0\cdot 2^2=0 \Rightarrow 0,1,2,3$

$1\cdot 2^2=4 \Rightarrow 4,5,6,7$

$2\cdot 2^2=8 \Rightarrow 8,9,10,11$

$\Rightarrow 12,13,14,15$

C \ D	0	1
0	0	0
1	1	1

0

C \ D	0	1
0	0	0
1	1	0

2

C \ D	0	1
0	1	1
1	0	1

1

C \ D	0	1
0	1	1
1	0	0

3

$\bar{B}C\bar{D}$

$\bar{A}CD$

$B(\bar{C}\bar{D}+\bar{C}D)$

$\Downarrow$

$=B\bar{C}(\underbrace{\bar{D}+D}_{1})$

$=B\bar{C}$

$f=\bar{B}C\bar{D}+\bar{A}CD+B\bar{C}$

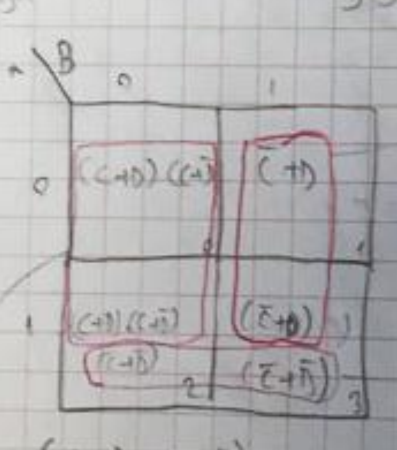
[illegible] 0'ları indirgeyerek

[illegible] kullanır

$B+(C+D)\cdot(C+\bar{D})$

$C\cdot C+C\bar{D}+CD+D\bar{D}$

$C\cdot(\underbrace{1+D+\bar{D}}_{1})=C$



$=(\bar{B}+\bar{C}+D)$

$=(\bar{A}+\bar{C}+\bar{D})$

$f=[B+(C+D)\cdot(C+\bar{D})]\cdot[\bar{B}+\bar{C}+D]\cdot$
$[\bar{A}+\bar{C}+\bar{D}]$

$f=(B+C)\cdot(\bar{B}+\bar{C}+D)\cdot(\bar{A}+\bar{C}+\bar{D})$

$B+(C+D)\cdot(C+\bar{D})$

Sor 3 öğretmen 3 sorudan 40, 40, 20 puanlık bir sınav kağıdını 100 üzerinden değerlendir. 1. ve 2. öğret- 40 puanlık soruları 3. öğretme ise 20 puanlık soruyu aşağıdaki şartlara göre değerlendiriyor

1. öğretmen : 0, 15, 25, 40 verir

2. öğret e : 0, 20, 30, 40

3. öğretme : 0, 20 (ya 0, ya da 20 verir)

Öğrencinin başarı durumunu gösteren devreyi yalnızca NAND kapılarını kullanarak gerçekleştir.

I. Tecr

A_1 A_0	
0 0	0
0 1	15
1 0	25
1 1	40

II. Tec

B_1 B_0	
0 0	0
0 1	20
1 0	30
1 1	40

C	
0	0
1	20

	Dc	A_1	A_0	B_1	B_0	C	f	
0	0	0	0	0	0	0	0	0
	1	0	0	0	0	1	0	
1	2	0	0	0	1	0	0	0
	3	0	0	0	1	1	0	
2	4	0	0	1	0	0	0	C
	5	0	0	1	0	1	1	
3	6	0	0	1	1	0	0	C
	7	0	0	1	1	0	1	
4	8	0	1	0	0	0	0	0
	9	0	1	0	0	1	0	
5	10	0	1	0	1	0	0	C
	11	0	1	0	1	1	1	
6	12	0	1	1	0	0	0	C
	13	0	1	1	0	1	1	
7	14	0	1	1	1	0	1	$C+\bar{C}$
	15	0	1	1	1	1	1	

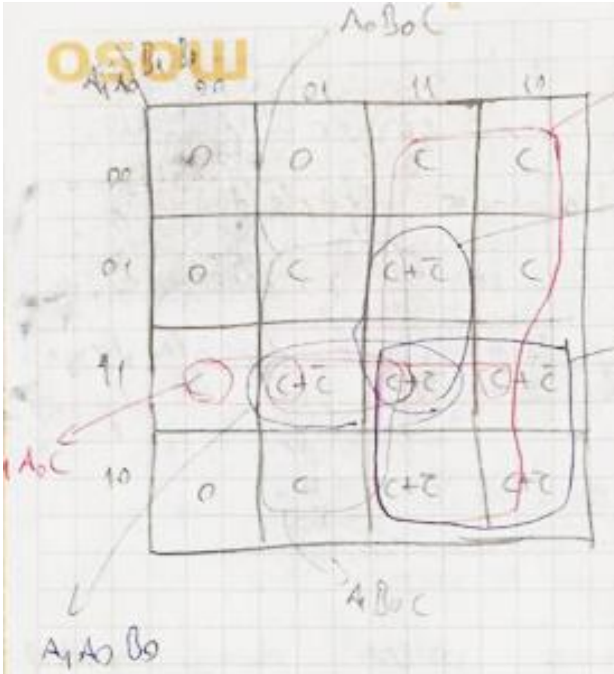
		A_1	A_0	B_1	B_0	C		
8	16	1	0	0	0	0	0	0
	17	1	0	0	0	1	0	
9	18	1	0	0	1	0	0	C
	19	1	0	0	1	1	1	
10	20	1	0	1	0	0	1	$C+\bar{C}$
	21	1	0	1	0	1	1	
11	22	1	0	1	1	0	1	$C+\bar{C}$
	23	1	0	1	1	1	1	
12	24	1	1	0	0	0	0	C
	25	1	1	0	0	1	1	
13	26	1	1	0	1	0	1	$C+\bar{C}$
	27	1	1	0	1	1	1	
14	28	1	1	1	0	0	1	$C+\bar{C}$
	29	1	1	1	0	1	1	
15	30	1	1	1	1	0	1	$C+\bar{C}$
	31	1	1	1	1	1	1	

$N=5$

$n=4$

$N-n=1$ (C)

$2^{N-n}=2^1=2$



$\rightarrow C B_1$

$\rightarrow A_0B_1B_0\underbrace{(C+\bar{C})}_{1}$

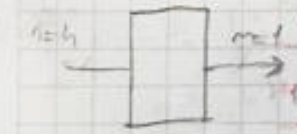
$(C+\bar{C})\cdot(A_1B_1)=A_1B_1$

[illegible]

Canonical POS → $f_3 = D\cdot\bar{D}$
POS → $f_3 = 0$

S- Bir hastahanede 5 tane asansör var 5. asansör yedek olarak kullanılıyor. 4 asansörden 2 veya daha fazlası meşgul olduğunda 5. asansör devreye giriyor. Böyle bir sisteme ait lojik devreyi yalnızca NOR kapılarını kullanarak yapınız.

Çıkışı 0 da aktif yapalım bu daha uygun



		A	B	C	D		
0	0	0	0	0	0	1	1
	1	0	0	0	1	1	
1	2	0	0	1	0	1	$\bar{D}$
	3	0	0	1	1	0	
2	4	0	1	0	0	1	$\bar{D}$
	5	0	1	0	1	0	
3	6	0	1	1	0	0	$D\cdot\bar{D}$
	7	0	1	1	1	0	
4	8	1	0	0	0	1	$\bar{D}$
	9	1	0	0	1	0	
5	10	1	0	1	0	0	$D\cdot\bar{D}$
	11	1	0	1	1	0	
6	12	1	1	0	0	0	$D\cdot\bar{D}$
	13	1	1	0	1	0	
7	14	1	1	1	0	0	$D\cdot\bar{D}$
	15	1	1	1	1	0	

Bunu 3 dereceden karnoş haritasıyla gerçekleştirin